

# CSNS RCS IPM — Electron-mode and ion-mode results

**Code:** Virtual-IPM 2.3.1. **Reference:** Rehman et al., NIM A **1092** (2026) 171809. **Cage:** 220 × 231 mm, design  $V = 25$  kV so  $E_y = V/d \approx 108$  kV/m; design  $B_y = 0.1$  T. **Statistics:** 100000 secondaries per run; quoted  $\sigma$  is the RMS of **detected** x. Space charge is the Gaussian bunch E plus Lorentz-boosted bunch B, compared with those fields off. Guiding E and B stay uniform. This report covers the **round-beam** e-mode replan (Blocks A-D + fine C/D) and the parallel **ion-mode** matrix (Blocks A-D). Elliptical 25×20 / 10×8 mm archives remain in REPORT.md §§1–6 as a closed reference; they are not reused here.

## 1. Setup

|                                | Injection                | Extraction        |
|--------------------------------|--------------------------|-------------------|
| Proton energy                  | 80 MeV                   | 1.6 GeV           |
| Bunch length $\sigma_t$        | 120 ns                   | 20 ns             |
| Round-beam $\sigma_x=\sigma_y$ | 25 mm (painted) or 10 mm | 10 mm             |
| Bunch population at 100 kW     | $7.8 \times 10^{12}$     | same, $\propto P$ |

- **Electron mode:** Voitkiv DDCS, GasType Hydrogen only (Virtual-IPM has no N<sub>2</sub>/H<sub>2</sub>O electron DDCS). Collection at the top electrode. Tracker: Boris, 8000 steps. RNG 1234.
- **Ion mode:** ZeroMomentum ions at rest masses 2 / 18 / 28 u (H<sub>2</sub><sup>+</sup>, H<sub>2</sub>O<sup>+</sup>, N<sub>2</sub><sup>+</sup> — the paper ToF peaks). Collection at the bottom electrode. Bunch fields off in the generator plus a circular 3-bunch train.
- **SC-off** is run once per geometry at 100 kW and shared across power (no bunch field  $\Rightarrow$  independent of bunch population).

Re-run: ./scripts/run\_emode\_replan.sh, ./scripts/run\_emode\_fine\_cd.sh, ./scripts/run\_imode\_replan.sh  
. Evaluate: python3 scripts/evaluate\_emode.py --from-summary, python3 scripts/evaluate\_imode.py --from-summary.

## 2. Takeaways

1. **Electron IPM is recoverable with guiding B.** Expansion oscillates with B (focusing / over-kick of opposite-sign electrons). **300 G** keeps every scanned round-beam family, power (100–500 kW), cage voltage (5–30 kV), and  $\Delta y$  offset ( $\pm 10$  mm) within  $\leq 1\%$ . Painted injection recovers first ( $\sim 105$ – $155$  G); extraction and 10 mm injection need  $\sim 190$ – $290$  G.
2. **Ion IPM is not recovered by B.** Expansion is flat from 0 to 200 G and only slightly lower at 0.1 T. Cyclotron radii at these fields are metres. **Do not use the e-mode B knob on ions.**
3. **Ion distortion scales with power and inversely with beam size.** H<sub>2</sub><sup>+</sup> injection, 10 mm, 0.1 T: +82% at 100 kW  $\rightarrow$  +422% at 500 kW. At  $\sigma=3$  mm / 500 kW the obtained width saturates on the cage ( $\sim 56$  mm, +1700%+) for all three species.
4. **Cage voltage has opposite roles.** In e-mode at B=0, 100 kW **crosses zero at 13 kV** (expansion  $\rightarrow$  compression); 500 kW never flips. In ion-mode expansion stays **positive** at every V; higher V shortens ToF and **reduces** the kick (H<sub>2</sub><sup>+</sup> inj. 10 mm @ 0.1 T: +198% at 5 kV  $\rightarrow$  +70% at 30 kV).
5. **Offset:**  $\Delta x$  is translation-invariant in both modes.  $\Delta y$  (toward/away from the detector) changes expansion by a few percent and does not undo the 300 G e-mode recovery.
6. **Practical implication:** tune B and V for a faithful **electron** profile; treat **ion-mode sizes as space-charge biased** unless a correction is applied.

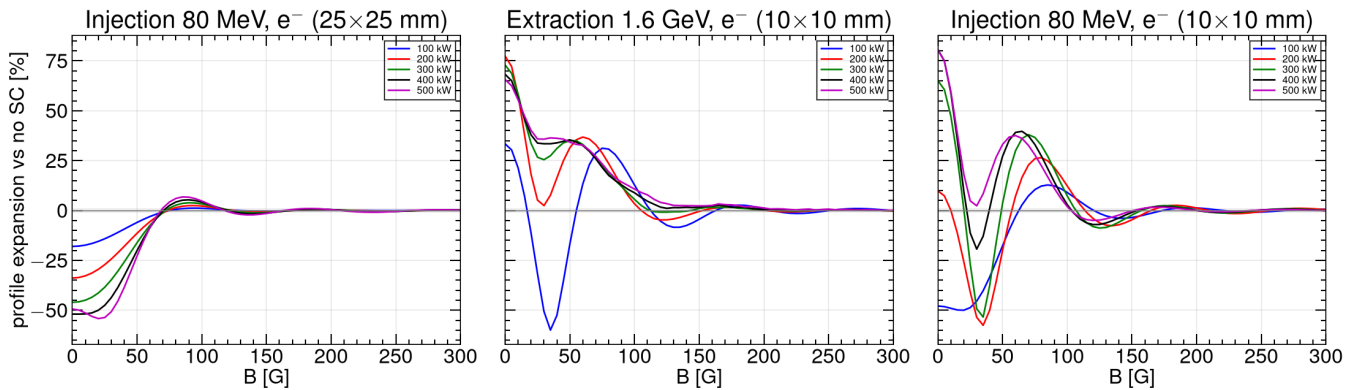
## 3. Electron mode

**Status: complete.** Replan A-D 2502 / 2502; fine C/D 3270 / 3270 new (5772 particle CSVs including A-D). All blocks use **round beams**  $\sigma_y = \sigma_x$ .

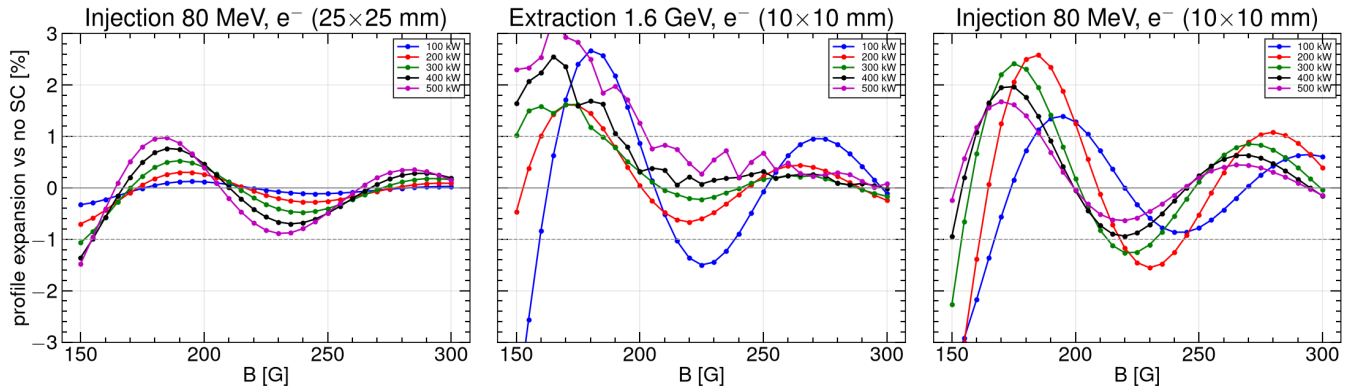
| Block  | What is scanned    | Beams                              | $\sigma$       | B                                | Power       |
|--------|--------------------|------------------------------------|----------------|----------------------------------|-------------|
| A      | Guiding field      | inj. 25 mm; ext. 10 mm; inj. 10 mm | round          | 0-300 G / 5 G                    | 100-500 kW  |
| B      | Beam size          | inj.; ext.                         | 3-20 mm / 1 mm | 0, 100, 200, 300 G, 0.1 T        | 100-500 kW  |
| C      | Cage voltage       | inj. 10 mm                         | round          | 0-300 G / 25 G + 0.1 T           | 100, 500 kW |
| Fine C | Voltage $\times$ B | inj. 10 mm                         | round          | diagnostic B + 5 G at 10-25 kV   | 100, 500 kW |
| D      | Beam offset        | inj.; ext. 10 mm                   | round          | 0, 100, 200, 300 G, 0.1 T        | 100, 500 kW |
| Fine D | $\Delta y$         | inj.; ext. 10 mm                   | round          | diagnostic B + 5 G at $\pm 5$ mm | 100, 500 kW |

Diagnostic B: 0, 25, 50, 75, 100, 125, 150, 200, 250, 300, 1000 G. Fine C voltages: 5-30 kV / 1 kV. Fine D:  $\Delta y = -10 \dots +10$  mm / 1 mm,  $\Delta x = 0$ .

### 3.1 Block A — B scan 0-300 G



**Figure 1.** Profile expansion vs no-SC for the three round reference beams, 100-500 kW. The grey band is  $\pm 1\%$ .



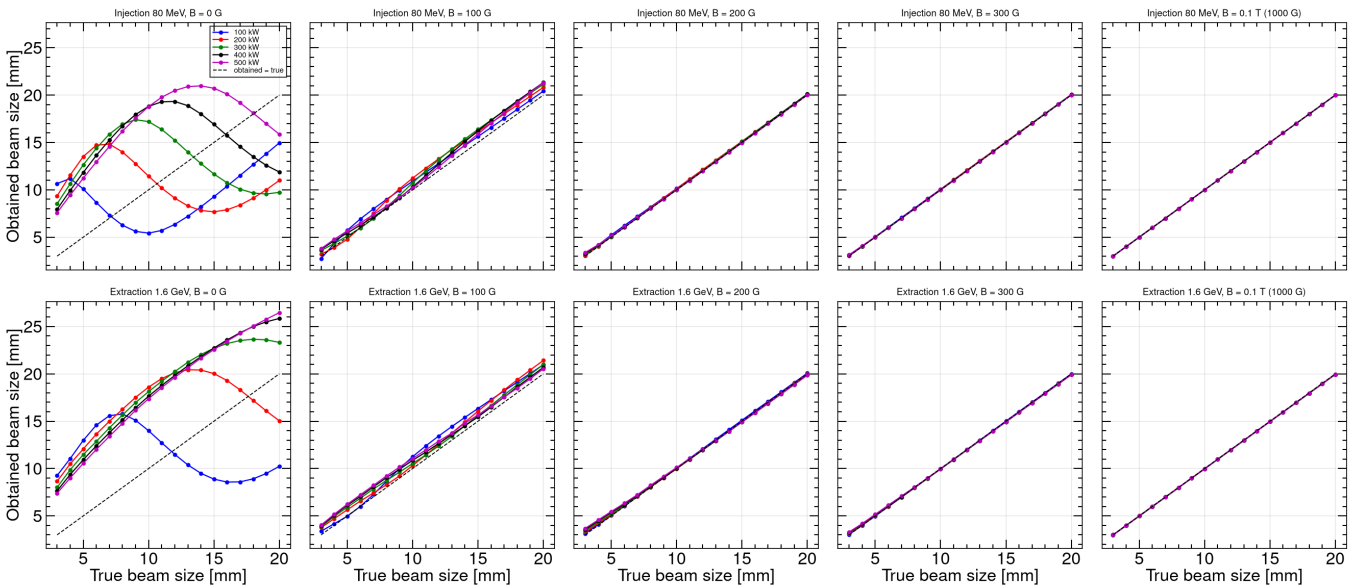
**Figure 2.** Same scan, 150–300 G,  $\pm 3\%$  zoom. Extraction and 10 mm injection still oscillate through  $\sim 150$ –250 G; **300 G** has settled.

Smallest B after which  $|\Delta|$  stays below 1% for the rest of the grid, and  $\Delta$  at 300 G:

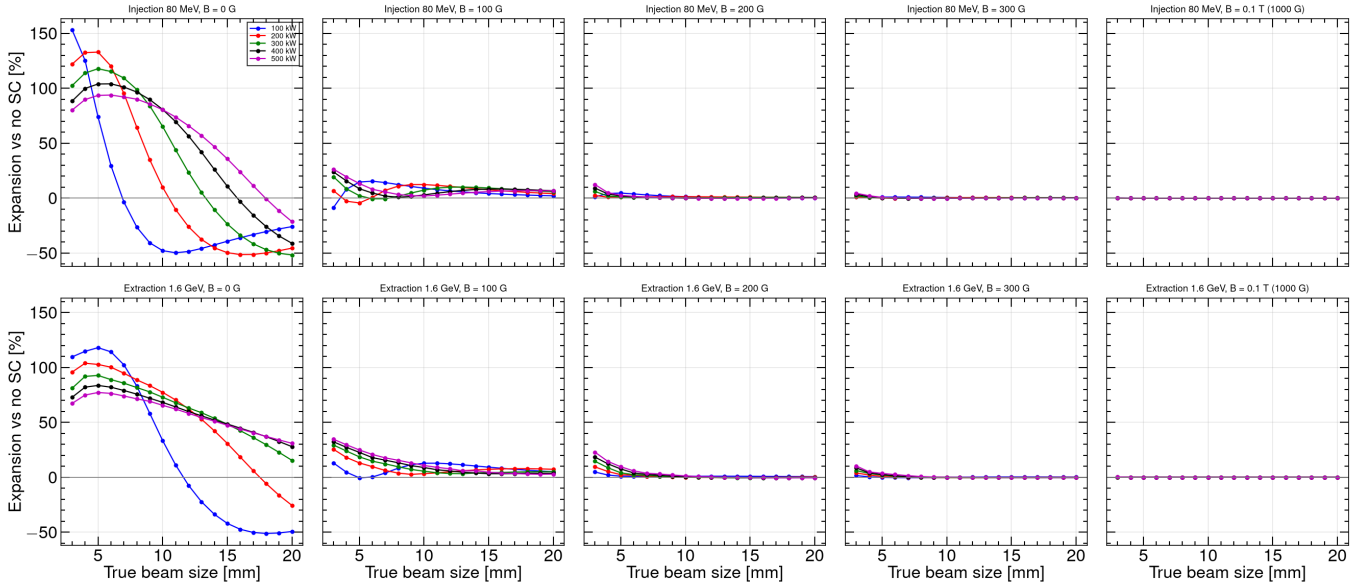
| Family        | 100 kW         | 200 kW         | 300 kW         | 400 kW         | 500 kW         |
|---------------|----------------|----------------|----------------|----------------|----------------|
| Inj. 25×25 mm | 105 G (+0.03%) | 115 G (+0.09%) | 155 G (+0.16%) | 155 G (+0.19%) | 155 G (+0.15%) |
| Ext. 10×10 mm | 240 G (−0.11%) | 190 G (−0.24%) | 185 G (−0.17%) | 195 G (−0.02%) | 205 G (+0.08%) |
| Inj. 10×10 mm | 210 G (+0.61%) | 290 G (+0.40%) | 235 G (−0.04%) | 190 G (−0.16%) | 190 G (−0.15%) |

Painted injection recovers first. The last curve to settle is 10×10 mm injection at 200 kW (threshold 290 G). **300 G keeps every power  $\leq 1\%$ .**

### 3.2 Block B — size 3-20 mm



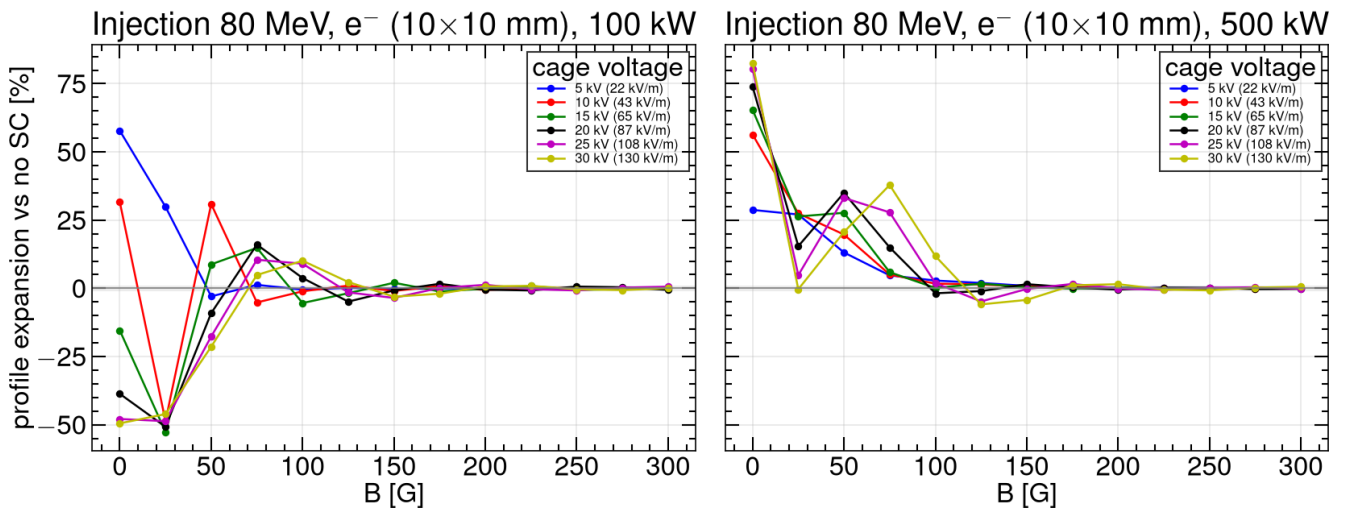
**Figure 3.** Obtained  $\sigma$  vs true  $\sigma$  at 0, 100, 200, 300 G and 0.1 T. Dashed line: obtained = true.



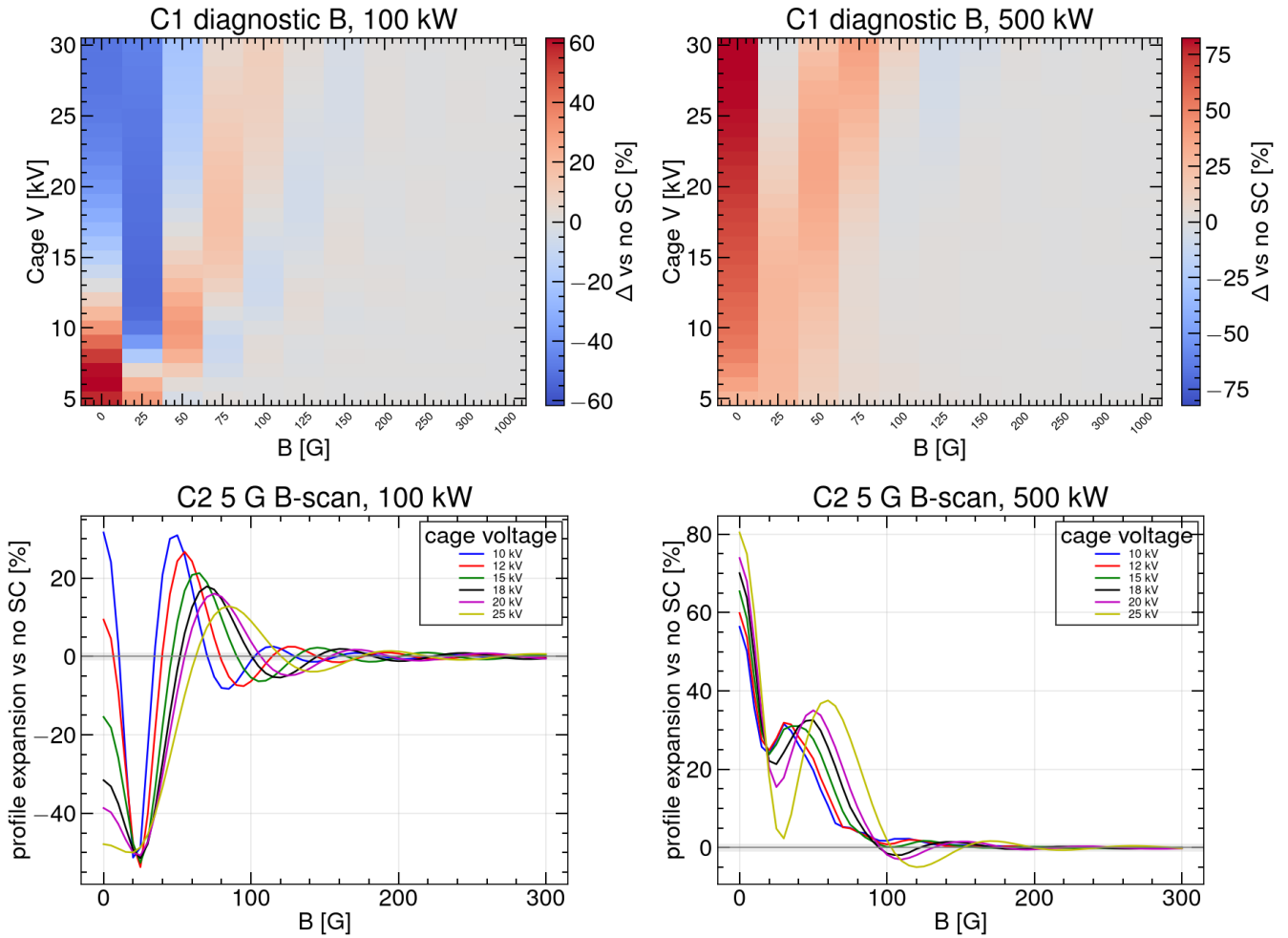
**Figure 4.** Expansion vs true beam size at the same B points.

At  $B=0$  obtained  $\sigma$  is **not** a monotonic function of true  $\sigma$  (focusing vs over-kick). From **200 G** the points lie on the diagonal except the tiniest beams. At **300 G**, injection 3 mm is still +1% (100 kW) to +4% (500 kW). **0.1 T** puts 3, 10, and 20 mm on the diagonal at every power.

### 3.3 Block C — cage voltage, including the 1 kV / 5 G fine scan



**Figure 5.** Coarse C: 5-30 kV / 5 kV on injection 10×10 mm.



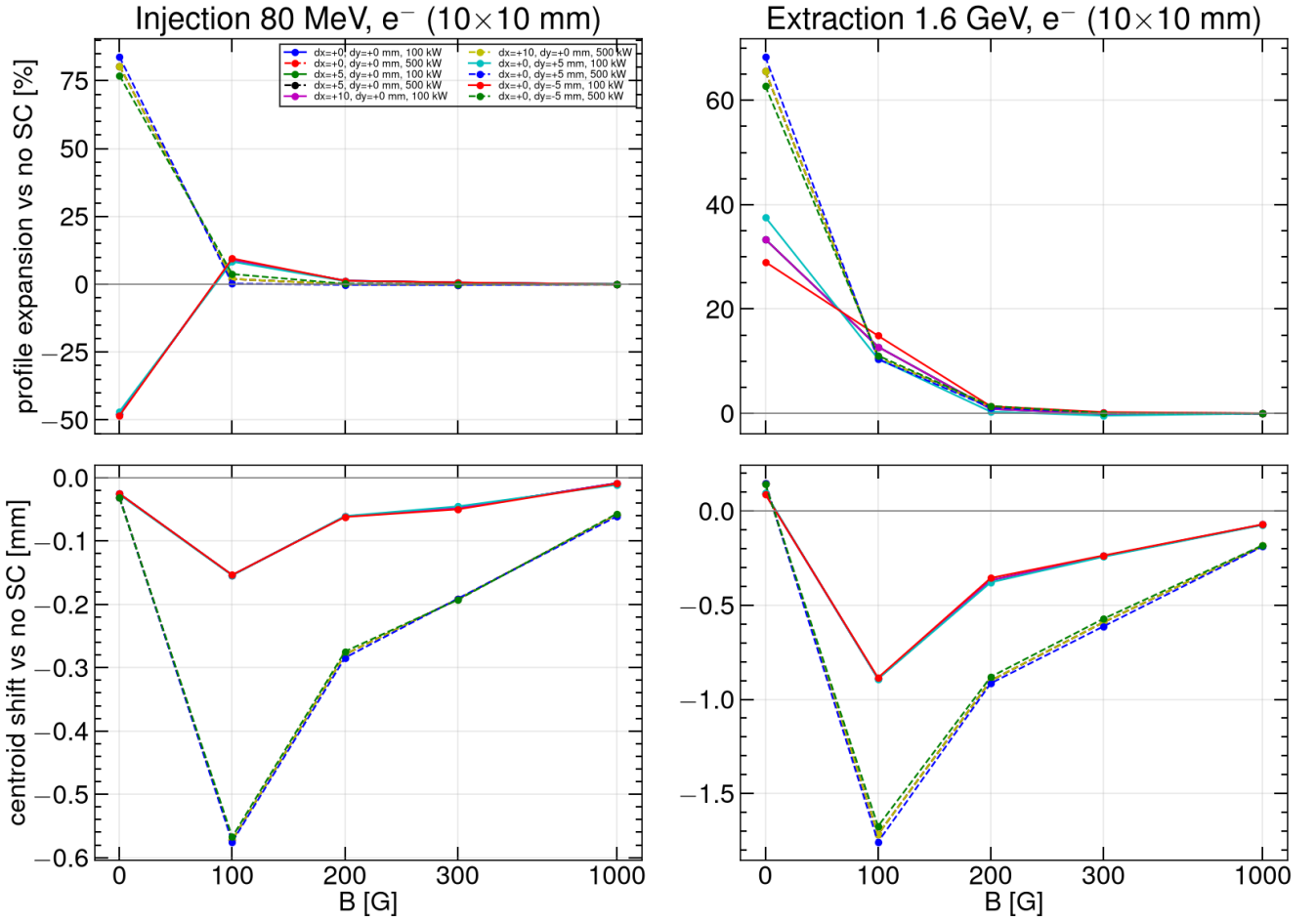
**Figure 6.** Fine C: 1 kV heatmap at diagnostic B, and 5 G B-scans at 10, 12, 15, 18, 20, 25 kV. At B=0, **100 kW crosses zero at 13 kV** (12 kV +9.4%, 13 kV ~0%, 14 kV –8.3%). **500 kW never flips:**  $\Delta$  stays positive and grows with V (+29% at 5 kV to +83% at 30 kV).

The 5 G scans show the same cyclotron/ToF oscillation as Block A, with the first peak later in B as V rises. 1% threshold on 0–300 G:

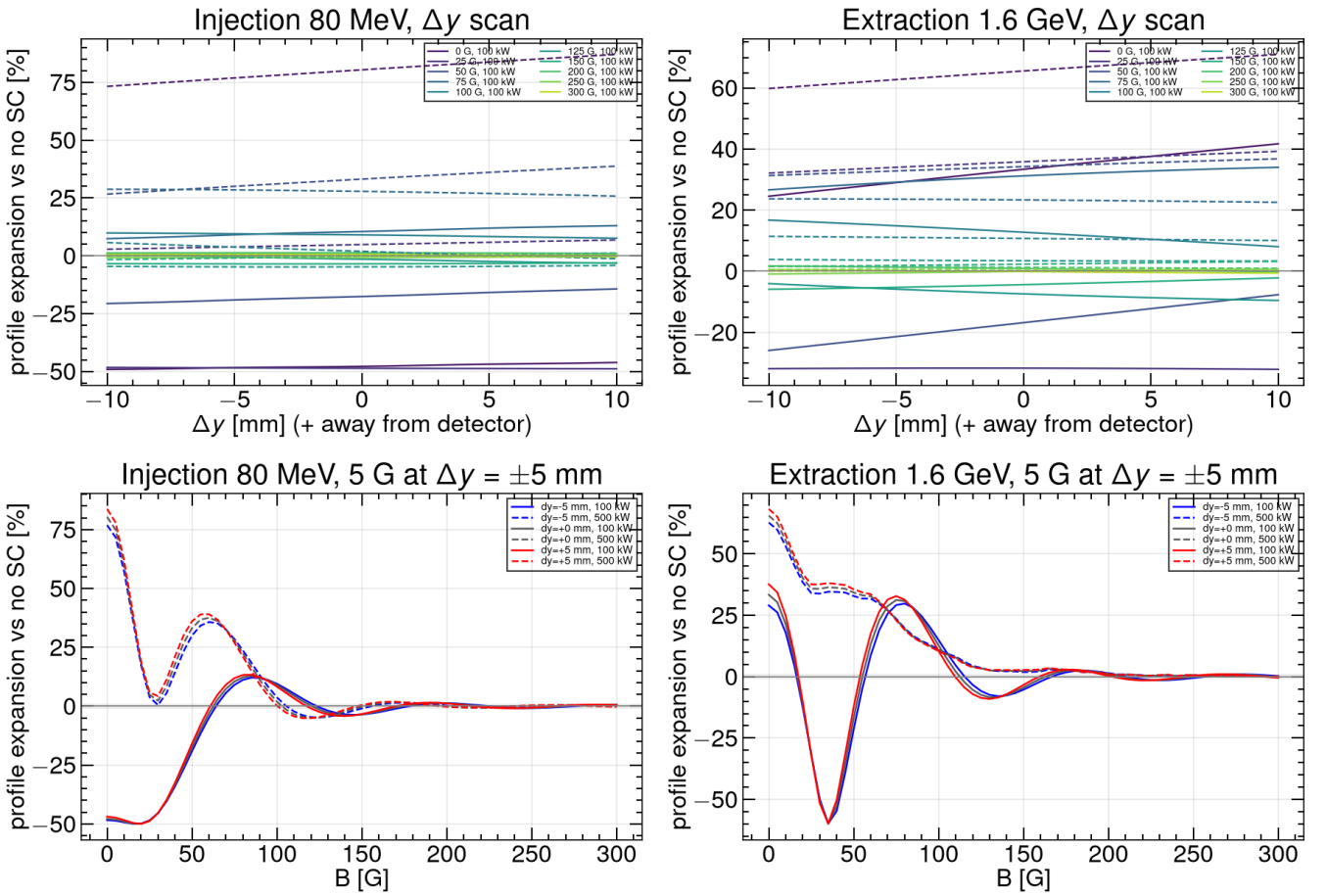
| V [kV]       | 100 kW | 500 kW | $\Delta$ at 300 G (100 / 500 kW) |
|--------------|--------|--------|----------------------------------|
| 10           | 150 G  | 130 G  | +0.15% / +0.02%                  |
| 12           | 195 G  | 135 G  | –0.31% / –0.05%                  |
| 15           | 195 G  | 145 G  | +0.47% / +0.03%                  |
| 18           | 210 G  | 160 G  | –0.47% / +0.01%                  |
| 20           | 220 G  | 170 G  | –0.57% / –0.19%                  |
| 25 (Block A) | 210 G  | 190 G  | +0.61% / –0.15%                  |

**300 G** still puts every scanned voltage  $\leq 1\%$ . Higher cage voltage needs more B at 100 kW; 500 kW recovers earlier at every V.

### 3.4 Block D — beam offset, including the 1 mm $\Delta y$ fine scan



**Figure 7.** Coarse D: four offsets on 10×10 mm injection and extraction.



**Figure 8.** Fine D: expansion vs  $\Delta y$  at diagnostic B, and 5 G B-scans at  $\Delta y = \pm 5$  mm vs centred. Expansion vs  $\Delta y$  is **smooth and nearly linear** from  $-10$  to  $+10$  mm. Toward the detector ( $\Delta y < 0$ ) vs away ( $\Delta y > 0$ ) changes  $\Delta$  by a few percent at  $B=0$  and by  $\leq 1\%$  once  $B=300$  G. The  $\pm 5$  mm 5 G scans overlay the centred Block A curves; the 1% threshold moves by at most  $\sim 10$  G.

| Family      | $\Delta y = -5$ mm | centred | $\Delta y = +5$ mm |
|-------------|--------------------|---------|--------------------|
| Inj. 100 kW | 210 G              | 210 G   | 205 G              |
| Inj. 500 kW | 190 G              | 190 G   | 185 G              |
| Ext. 100 kW | 245 G              | 240 G   | 235 G              |
| Ext. 500 kW | 205 G              | 205 G   | 205 G              |

At **300 G** and **0.1 T**, expansion stays  $\leq 1\%$  for every offset. Centroid shift vs no-SC is  $\leq 0.2$  mm (injection) and  $\leq 0.6$  mm (extraction, 500 kW, 300 G). A few-mm orbit offset does not undo the B recovery. No  $\Delta x$  scan: the cage is translation-invariant in x.

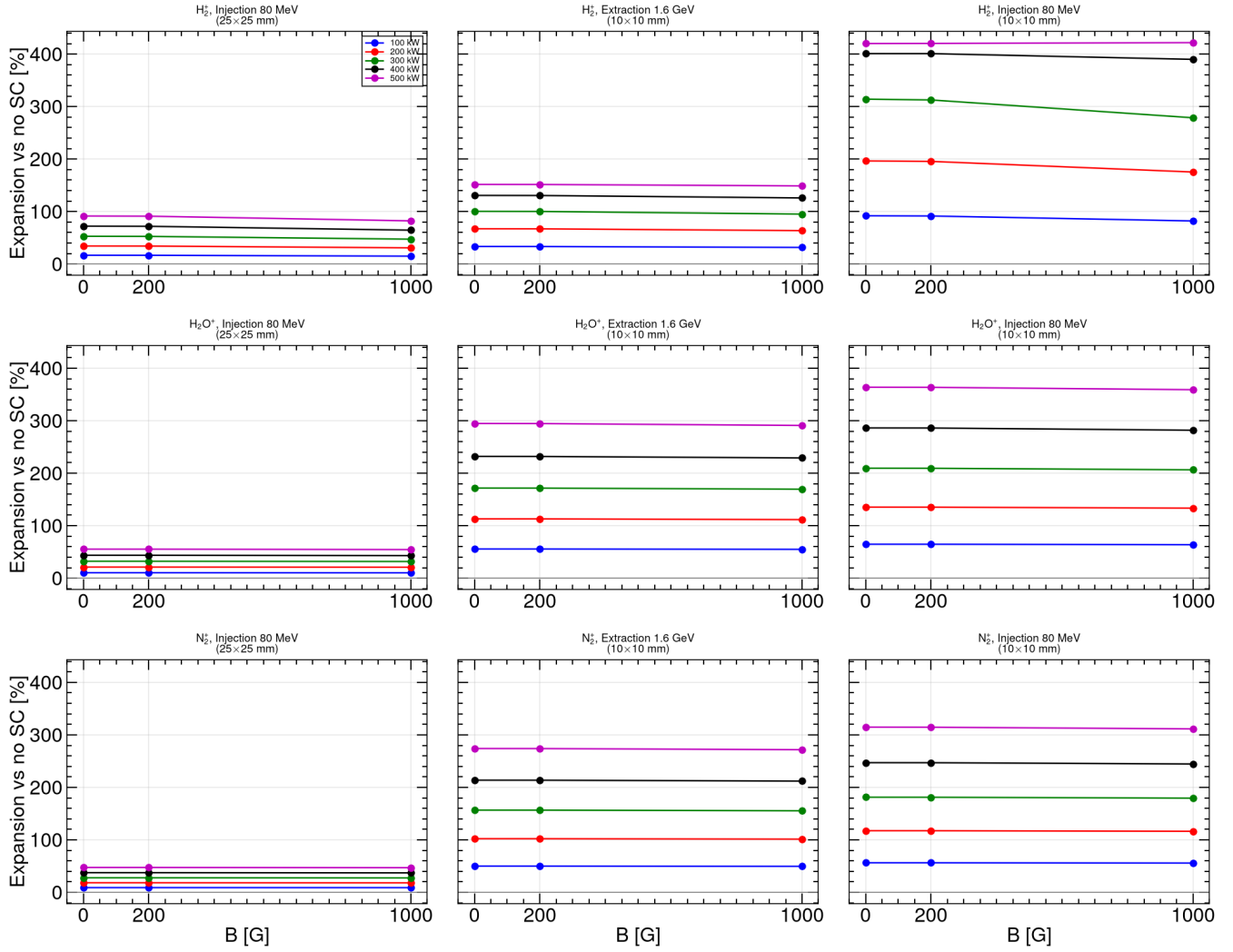
## 4. Ion mode

**Status: complete.** 2070 / 2070 runs. Round beams;  $H_2^+$  /  $H_2O^+$  /  $N_2^+$ . **No dense B-scan** — only 0, 200, 1000 G — because closed ion scans were already flat vs B.

| Block | What is scanned | Beams                              | $\sigma$       | B [G]        | Power [kW] |
|-------|-----------------|------------------------------------|----------------|--------------|------------|
| A     | Reference beams | inj. 25 mm; ext. 10 mm; inj. 10 mm | round          | 0, 200, 1000 | 100-500    |
| B     | Beam size       | inj.; ext.                         | 3-20 mm / 1 mm | 0, 200, 1000 | 100-500    |
| C     | Cage voltage    | inj. 10 mm                         | round          | 0, 200, 1000 | 100, 500   |
| D     | Beam offset     | inj.; ext. 10 mm                   | round          | 0, 200, 1000 | 100, 500   |

SC-off reference:  $H_2^+$  @ 100 kW only (mass-independent without bunch fields, shared).

### 4.1 Block A — reference beams vs B



**Figure 9.** Expansion vs B for the three reference beams. Each row is a species; columns are painted injection (25×25 mm), extraction (10×10 mm), and small injection (10×10 mm).

Expansion is **essentially independent of B** between 0 and 200 G. The 0.1 T point is sometimes slightly lower but never restores the profile.

Expansion at 100 kW [%]:

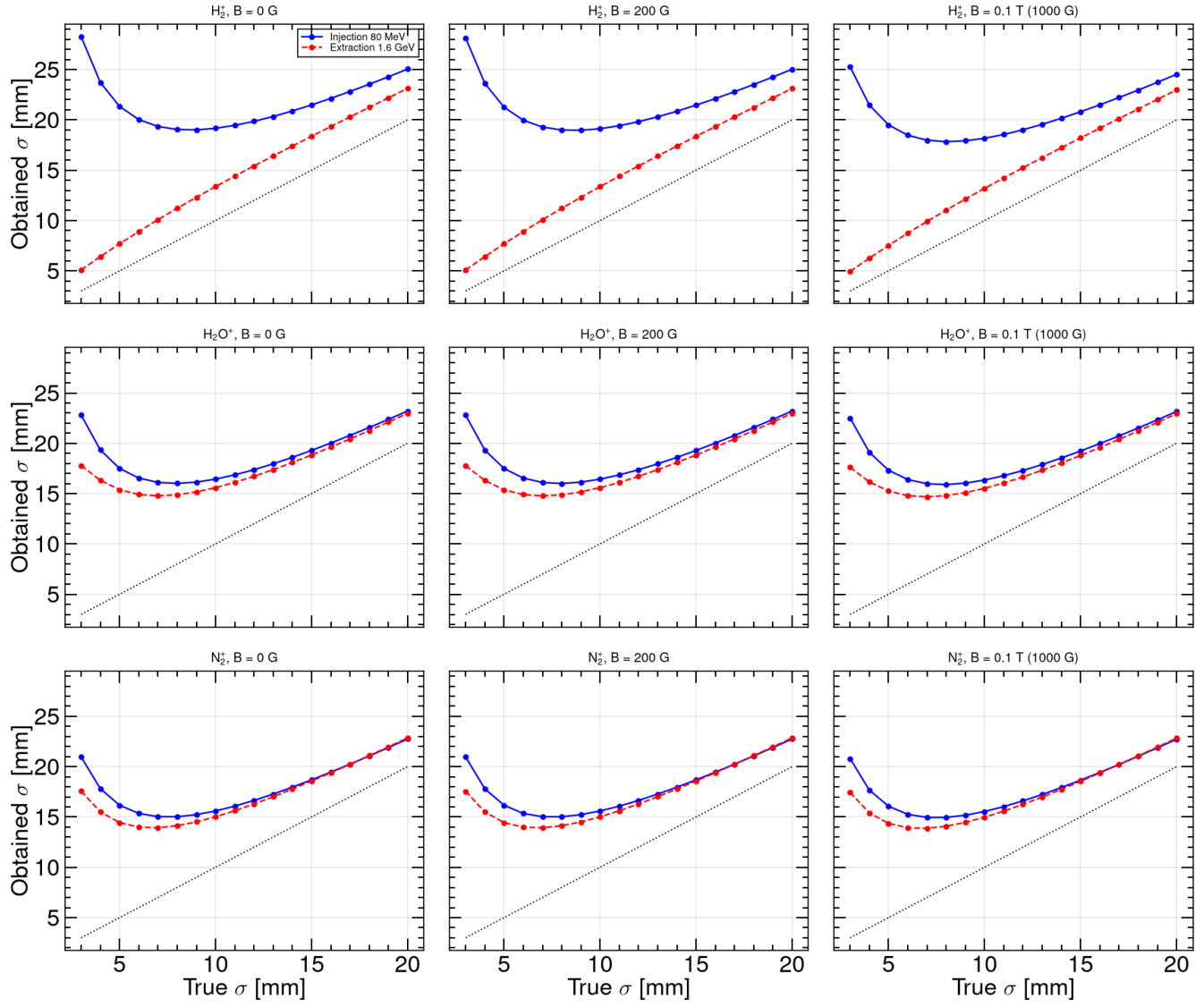
| Species                       | Inj. 25 mm 0 G | 200 G | 1000 G | Inj. 10 mm 0 G | 200 G | 1000 G | Ext. 10 mm 0 G | 200 G | 1000 G |
|-------------------------------|----------------|-------|--------|----------------|-------|--------|----------------|-------|--------|
| H <sub>2</sub> <sup>+</sup>   | +16.7          | +16.7 | +15.0  | +92.0          | +91.6 | +82.1  | +33.5          | +33.4 | +31.6  |
| H <sub>2</sub> O <sup>+</sup> | +10.4          | +10.4 | +10.2  | +64.7          | +64.7 | +63.8  | +55.6          | +55.6 | +54.9  |
| N <sub>2</sub> <sup>+</sup>   | +8.9           | +8.9  | +8.9   | +56.3          | +56.2 | +55.7  | +49.9          | +49.9 | +49.5  |

Painted injection matches the elliptical-beam design-field numbers ( $\sim +16\%$  H<sub>2</sub><sup>+</sup>). The 10 mm injection family is the most distorted at every B.

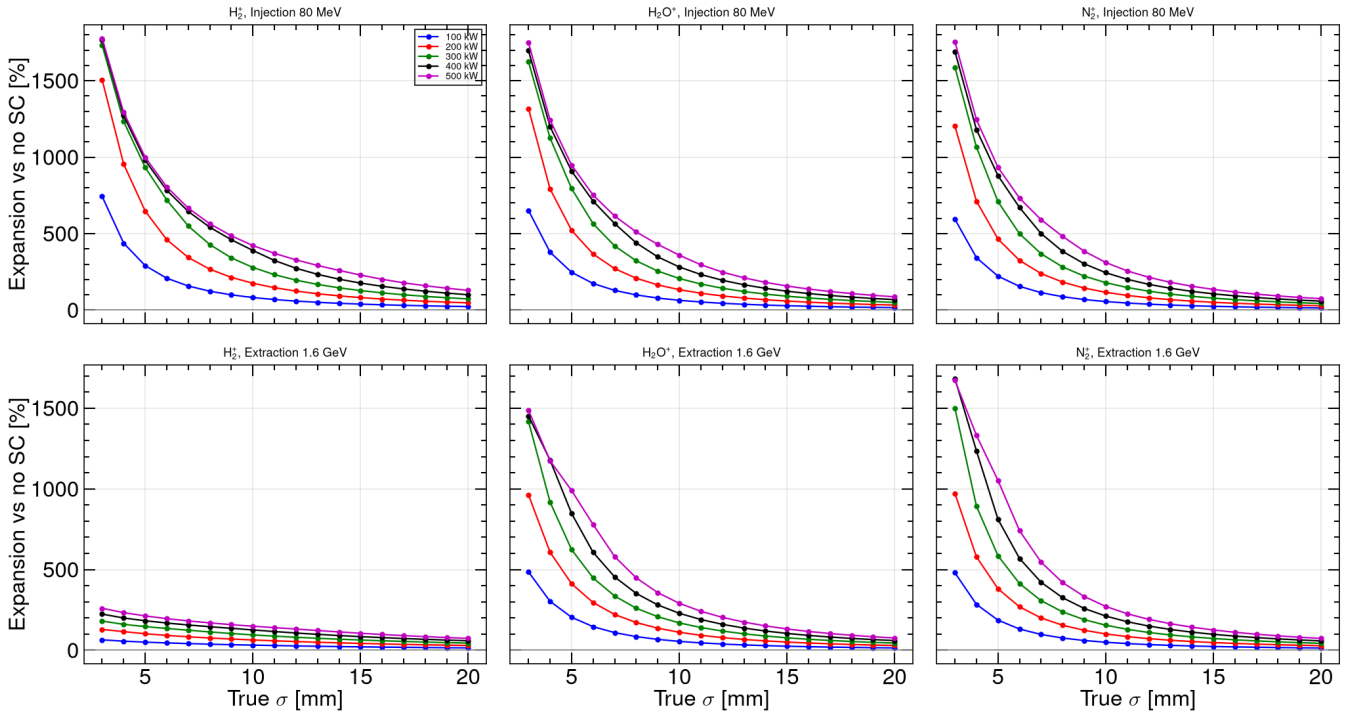
Extraction H<sub>2</sub><sup>+</sup> (+33.5 % at 0 G) is a **lower bound**. Ions are generated 125 ns before the first field-carrying bunch, so H<sub>2</sub><sup>+</sup> has already drifted  $\approx 40$  mm when the kick arrives. A line-charge model with aligned generation gives  $\approx +100\%$  (0 G, 100 kW). H<sub>2</sub>O<sup>+</sup> / N<sub>2</sub><sup>+</sup> barely move in 125 ns and are close to the whole-bunch limit. Injection is correctly aligned (both centres at 480 ns). Details and the correction recipe: LITERATURE\_REVIEW.md §§3.5–3.6.

## 4.2 Block B — size 3-20 mm





**Figure 10.** Obtained  $\sigma$  vs true  $\sigma$  at 0.1 T, 100 kW. Dashed line: obtained = true. Every point lies **above** the diagonal.



**Figure 11.** Expansion vs true  $\sigma$  at 0.1 T for 100–500 kW. Smaller beams and higher power inflate more.

Obtained size at  $\sigma_x = 10$  mm, 0.1 T, 100 kW:

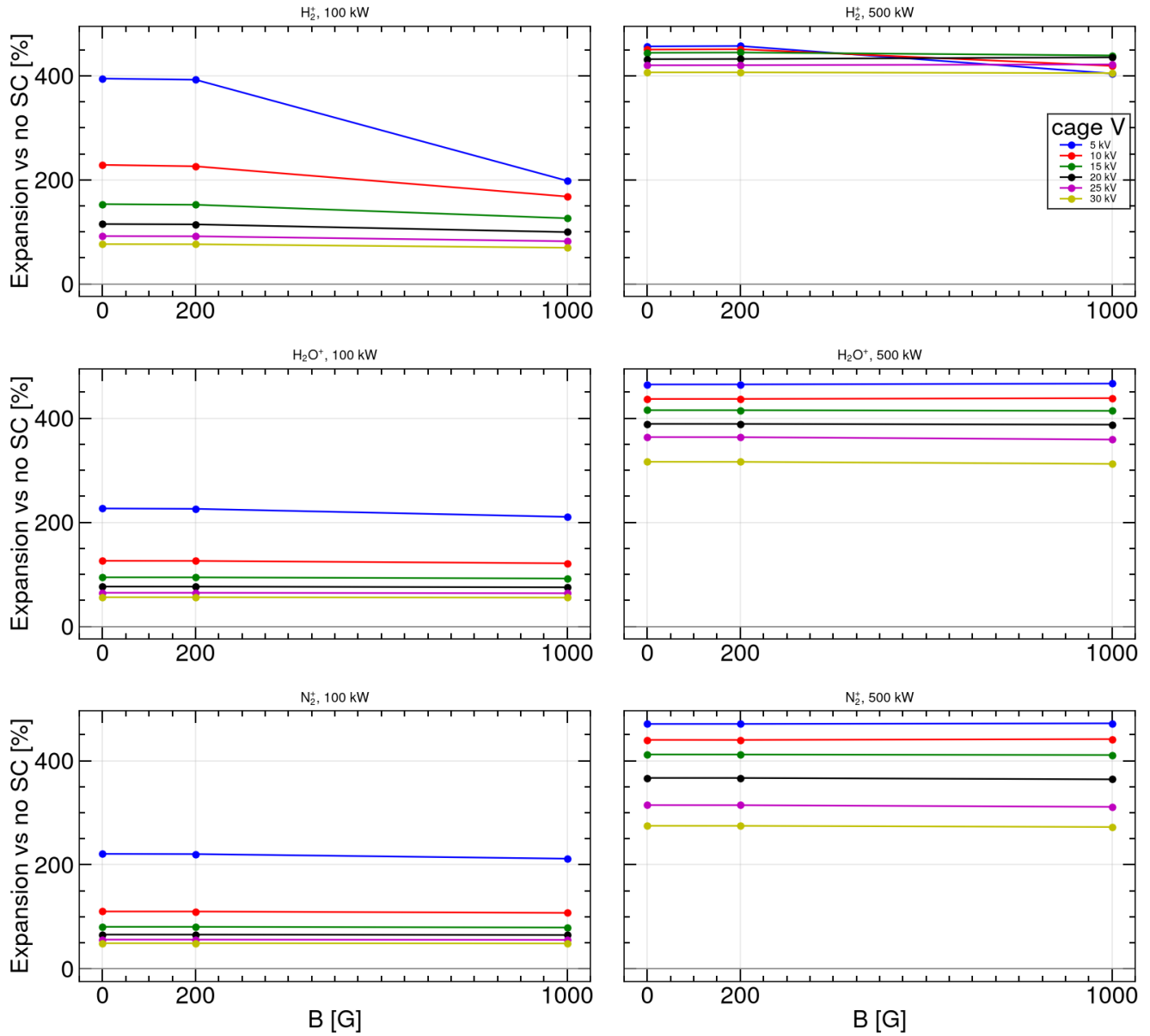
| Species  | Inj. obtained | Inj. expansion | Ext. obtained | Ext. expansion |
|----------|---------------|----------------|---------------|----------------|
| $H_2^+$  | 18.2 mm       | +82%           | 13.2 mm       | +32%           |
| $H_2O^+$ | 16.4 mm       | +64%           | 15.5 mm       | +55%           |
| $N_2^+$  | 15.5 mm       | +56%           | 15.0 mm       | +50%           |

$H_2^+$  injection,  $\sigma=10$  mm, 0.1 T vs power:

| 100 kW         | 200 kW          | 300 kW          | 400 kW          | 500 kW          |
|----------------|-----------------|-----------------|-----------------|-----------------|
| +82% (18.2 mm) | +175% (27.5 mm) | +279% (37.8 mm) | +390% (48.9 mm) | +422% (52.1 mm) |

Lighter ions expand more at injection ( $\Delta v = qEt E_{sc} dt/m$ ). At extraction the short 1.6 GeV bunch plus 0.1 T keeps  $H_2^+$  nearer the diagonal;  $H_2O^+$  and  $N_2^+$  stay in the cage longer and can inflate more. Worst case: **injection,  $\sigma=3$  mm, 500 kW, 0.1 T** — obtained  $\sigma \approx 56$  mm (+1700%+) for all three species (cage-wall saturation).

### 4.3 Block C — cage voltage 5–30 kV



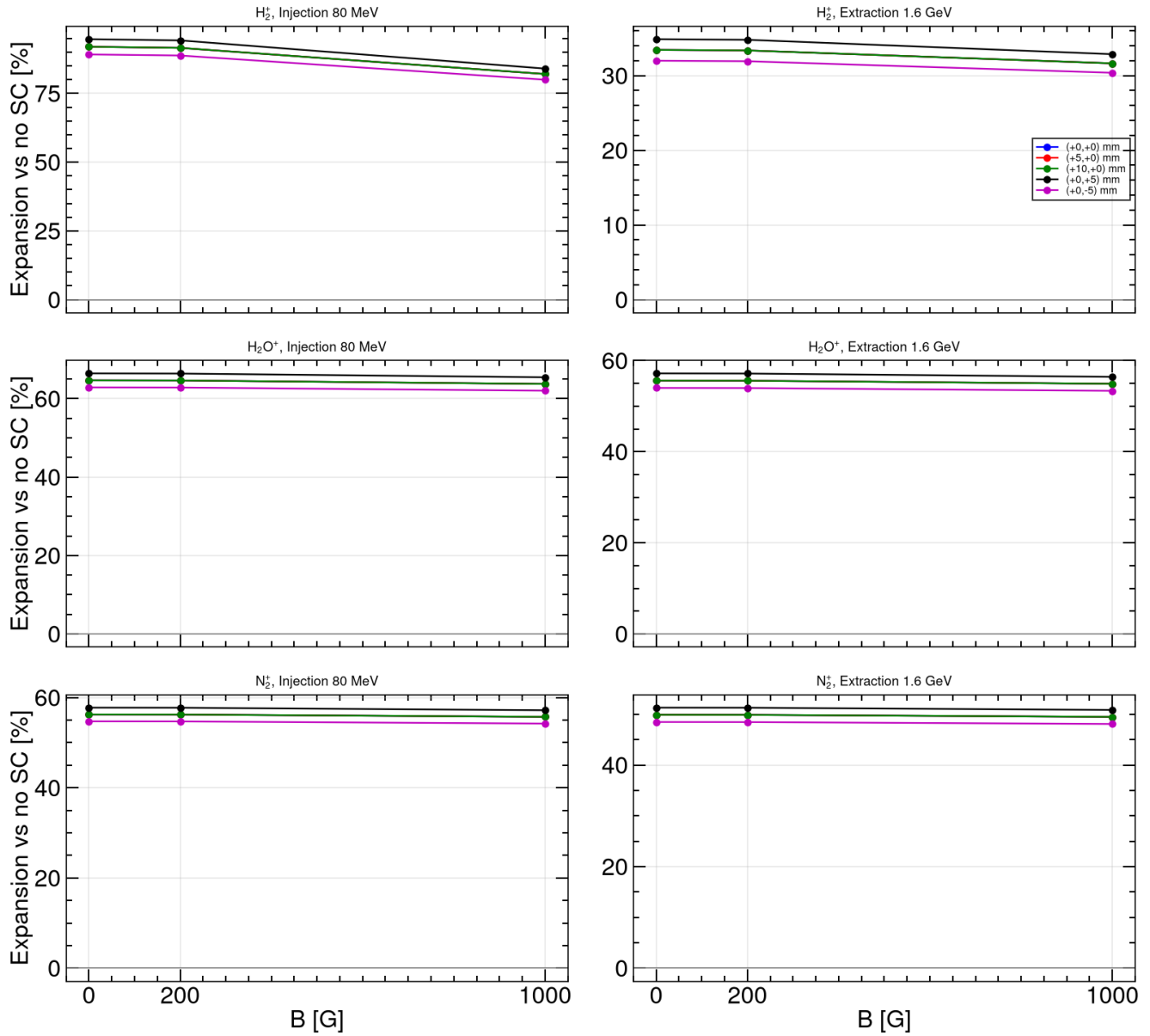
**Figure 12.** Expansion vs B for each cage voltage, injection  $10 \times 10$  mm. Unlike e-mode, ion expansion **never changes sign**.

$H_2^+$  injection 10 mm, 100 kW, 0.1 T vs cage voltage:

| 5 kV  | 10 kV | 15 kV | 20 kV | 25 kV | 30 kV |
|-------|-------|-------|-------|-------|-------|
| +198% | +168% | +126% | +100% | +82%  | +70%  |

Higher V shortens ion drift and reduces the integrated kick. At 500 kW the same trend holds at much larger expansion (+300% to +800% range).

#### 4.4 Block D — beam offset



**Figure 13.** Expansion vs B for each offset (100 kW).  $\Delta x=+5$  and  $+10$  mm overlay the centred beam.  $\Delta y=\pm 5$  mm changes expansion by a few percent.

$H_2^+$  injection 10 mm, 0.1 T, 100 kW vs offset:

| centred | $\Delta x=+5$ | $\Delta x=+10$ | $\Delta y=+5$ | $\Delta y=-5$ |
|---------|---------------|----------------|---------------|---------------|
| +82.1%  | +82.1%        | +82.1%         | +84.0%        | +80.0%        |

Toward the detector ( $\Delta y < 0$ ): slightly less expansion. Centroid shift vs no-SC is  $\lesssim 0.6$  mm at extraction, 500 kW.

## 5. Electron vs ion

|              | Electron mode                                           | Ion mode                               |
|--------------|---------------------------------------------------------|----------------------------------------|
| Secondaries  | $e^-$ (Voitkiv $H_2$ )                                  | $H_2^+ / H_2O^+ / N_2^+$               |
| B grid       | dense (5 G / 25 G)                                      | 0, 200, 1000 G only                    |
| Effect of B  | oscillatory recovery; 300 G fixes the profile           | flat; no recovery                      |
| Cage voltage | sign flip at 13 kV (100 kW, $B=0$ ); 500 kW never flips | always positive; higher V reduces kick |
| Beam offset  | $\Delta x$ invariant; $\Delta y$ few %                  | same                                   |
| Worst case   | low B, wrong V polarity                                 | small $\sigma$ , high P, injection     |

|              |                            |                                       |
|--------------|----------------------------|---------------------------------------|
| Design 0.1 T | conservative for electrons | still +15% to +80% (10–25 mm, 100 kW) |
|--------------|----------------------------|---------------------------------------|

**Operate the CSNS RCS IPM in electron mode with  $B \geq 300$  G.** Ion-mode profiles remain space-charge inflated at every scanned B, V, size, power, and offset; use them only with a correction model, or restrict ion ToF to species identification rather than size.

## 6. Data products

| File                                         | Contents                                                                  |
|----------------------------------------------|---------------------------------------------------------------------------|
| `output/csns_emode_bscan300_summary.csv`     | E-mode Block A                                                            |
| `output/csns_emode_size_summary.csv`         | E-mode Block B                                                            |
| `output/csns_emode_voltage_summary.csv`      | E-mode Block C (coarse)                                                   |
| `output/csns_emode_offset_summary.csv`       | E-mode Block D (coarse)                                                   |
| `output/csns_emode_voltage_fine_summary.csv` | Fine C                                                                    |
| `output/csns_emode_offset_fine_summary.csv`  | Fine D                                                                    |
| `output/csns_imode_bscan_summary.csv`        | Ion-mode Block A (135 rows)                                               |
| `output/csns_imode_size_summary.csv`         | Ion-mode Block B (1620 rows)                                              |
| `output/csns_imode_voltage_summary.csv`      | Ion-mode Block C (108 rows)                                               |
| `output/csns_imode_offset_summary.csv`       | Ion-mode Block D (180 rows)                                               |
| `output/csns_imode_kick_model.csv`           | Reduced line-charge model vs Virtual-IPM (0 G); aligned extraction column |
| `plots/csns_emode_*.png`                     | Figures 1–8                                                               |
| `plots/csns_imode_*.png`                     | Figures 9–13                                                              |

Particle CSVs under output/emode/ and output/imode/ are gitignored. Lab notebook with the closed elliptical campaigns: REPORT.md. Ion-only write-up: IMODE\_REPORT.md. Comparison with IPM papers 2006–2026: LITERATURE\_REVIEW.md / LITERATURE\_REVIEW.pdf.  
 Rebuild this PDF: python3 scripts/md\_to\_pdf.py --md CSNS\_IPM\_REPORT.md --pdf CSNS\_IPM\_REPORT.pdf --footer "CSNS RCS IPM — e-mode and ion-mode".